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TRANSPORT PHENOMENA II, CHE 312

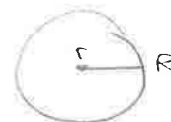
EXAM 2

CLOSED BOOK, TIME: 75 MINUTES

Solid species A in the shape of a sphere of radius R is submerged in a large body of stagnant liquid B. Species A is slightly soluble in the liquid B with an equilibrium concentration of c_{A0} at $r = R$. The dissolved species A diffuses radially in the liquid phase while it undergoes a first-order chemical reaction.

Answer the following questions in terms of the following constants: radius of the solid sphere R, the diffusivity D_{AB} , and the rate constant k_1''' .

- Using a shell mole balance for species A, derive an ordinary differential equation for the steady-state concentration profile of species A in the liquid phase. Assume $x_A \ll 1$
- Solve the above differential equation and express the concentration profile of species A in terms of the Thiele modulus $\phi \equiv \sqrt{\frac{k_1'''}{D_{AB}}} R$ and two integration constants. Hint: $\frac{1}{r} \frac{d}{dr} \left(r^2 \frac{dc_A}{dr} \right) = \frac{d^2}{dr^2} (rc_A)$
- Determine the integration constants using two appropriate boundary conditions for the concentration profile.
- Determine how many moles of A react per unit time.



$x_A \ll 1$ radius r , rc_A

$$1) N_{Ar} = \underbrace{c_A v_r}_{\text{No convection}} - c_A D_{AB} \frac{dx_A}{dr}$$

b/c $x_A \ll 1$

$$R_1 = k_1''' c_A \quad \text{reaction rate}$$

$$N_{Ar} = -D_{AB} \frac{dc_A}{dr}$$

overall mole balance:

$$-\int_V dv \frac{dc_A}{dt} = -\int_S ds \langle N_{Ar} \rangle + \int_V dv R_A$$

@ steady state

$$[4\pi r^2 N_{Ar}]_r - [4\pi r^2 N_{Ar}]_{r+\Delta r} + [4\pi r^2 \Delta r k_1''' c_A] = 0$$

Divide by $4\pi r^2 \Delta r$ & Take $\lim_{\Delta r \rightarrow 0}$

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$$\frac{1}{r^2} \frac{d}{dr} (r^2 N_A r) + K_1''' CA = 0$$

$$+ \frac{1}{r^2} \frac{d}{dr} (r^2 \underbrace{D_{AB}}_{\text{constant}} \frac{d}{dr} CA) = + K_1''' CA$$

$$\frac{1}{r^2} \frac{d}{dr} (r^2 \frac{d}{dr} CA) = \frac{K_1''' CA}{D_{AB}}$$

rewrite it in a different form:-

$$\boxed{\frac{1}{r} \frac{d}{dr} (r^2 \frac{d}{dr} CA)} = \frac{K_1''' CA r}{D_{AB}}$$

$$\frac{d^2}{dr^2} r CA = \frac{K_1''' CA r}{D_{AB}} \quad \left. \vphantom{\frac{d^2}{dr^2} r CA} \right\} \text{ordinary differential equation}$$

2) To solve it we need to do dimensionless

$$T = \frac{CA}{CA_0} \quad \xi = \frac{r}{R} \quad \phi = \sqrt{\frac{K_1'''}{D_{AB}}} \rightarrow \phi^2 = \frac{K_1'''}{D_{AB}}$$

$$\frac{d}{dr} = \frac{d}{d\xi} \frac{d\xi}{dr} = \frac{d}{d\xi} \frac{1}{R}$$

$$\frac{d^2}{dr^2} = \frac{d^2}{d\xi^2} \frac{1}{R^2}$$

$$\frac{1}{R^2} \frac{d^2}{d\xi^2} \cancel{R} \xi \cancel{CA_0 T} = \phi^2 \xi \cancel{R} \cancel{CA_0 T}$$

$$\frac{d^2}{d\xi^2} \xi T - \phi^2 \xi T = 0$$

using hyperbolic solution:

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$$\Gamma = \frac{\sin(\phi \xi)}{\xi \sin(\phi)}$$

$$C_A = C_{A0} \frac{\sin(\phi \frac{r}{R})}{\frac{r}{R} \sin(\phi)}$$

3) Boundary Conditions: $C_A(r=0) = 0$
 $C_A(r=R) = C_{A0}$

we will need to switch the boundary conditions to be in terms of ξ ! Γ ~~is~~
and we should integrate twice
Then apply them to find the constants.

4) $N_A r = -D_{AB} \frac{dC_A}{dr}$

$$\frac{dC_A}{dr} = \frac{dC_A}{d\xi} \frac{d\xi}{dr} = \frac{dC_A}{d\xi} \frac{1}{R}$$

and the rate would be:

$$4\pi R^2 N_A r$$

$$-\frac{4\pi R^2 D_{AB}}{R} \frac{dC_A}{d\xi} = -4\pi R D_{AB} (1 - \phi \coth \phi)$$